AI Research Report

Query: what is the latest research on residual learning

(F(xi)) (4) ReZero Net. xi+1 = xi + αi σ(F(xi)) (5) ReZero Net. + Norm xi+1 = xi + αi Norm (F(xi)) [...].

URL: https://proceedings.mlr.press/v161/bachlechner21a/bachlechner21a.pdf

---

SOURCE ID: [S7]

Title: [PDF] A Mean-field Analysis of Deep ResNet and Beyond

Content:

Samuel J. Yang, Lexing Ying. 2020. A mean-field analysis of deep ResNet and beyond. Journal of Machine Learning Research, 21(1):1–55.

Abstract: Deep residual networks (ResNets) have been extremely successful in various applications, but a rigorous theoretical understanding of their training dynamics remains lacking. In this paper, we propose a mean-■eld framework for analyzing deep ResNets, which allows us to study the training dynamics of these networks in a principled way. We derive a recursive formula for the evolution of the neural network parameters, and use it to analyze the training behavior of deep ResNets. Our analysis provides new insights into the training dynamics of deep ResNets, including the emergence of ■xed points, the role of regularization, and the e■ect of width and depth on training.

URL: https://web.stanford.edu/~lexing/mfresnet.pdf

SOURCE ID: [S8]

Title: Residual Networks Behave Like Ensembles of Relatively ...

Content:

In this paper, we analyze the behavior of residual networks, showing that they tend to behave like ensembles of relatively shallow networks. To do this, we consider the training dynamics of residual networks, and show that the learned representations are highly correlated across different depths of the network. This correlation allows us to approximate a deep residual network with a collection of relatively shallow networks, which leads to a better understanding of the training dynamics of these models.

URL: https://proceedings.neurips.cc/paper_files/paper/6556-residual-networks-behave-like-ensembles-of-relatively-shallow-networks.pdf

---

Research on Residual Learning

Introduction

Residual learning is a concept in deep learning that has gained significant attention in recent years [S1]. It involves using residual connections to improve the training of deep neural networks [S2]. Residual connections allow the output of a layer to be added to the input of the next layer, which helps to prevent vanishing gradients and improves the stability of the network [S3].

Key Findings

The latest research papers on residual learning have shown that it can be used to improve the performance of various deep learning models [S1]. Residual connections have been used in Transformer architectures, where they have been shown to improve the stability of the network and reduce the risk of gradient vanishing

[S4]. Pre-Norm and Post-Norm variants of residual connections have been proposed, which make distinct trade-offs between gradient vanishing and representation collapse [S4]. ReZero, a new residual connection method, has been proposed, which facilitates dynamical isometry and enables the efficient training of extremely deep networks [S5][S6].

Detailed Explanation

Research has shown that residual connections can be used to improve the performance of deep neural networks [S2]. The use of residual connections allows the output of a layer to be added to the input of the next layer, which helps to prevent vanishing gradients and improves the stability of the network [S3]. The mean-field analysis of deep ResNet has provided new insights into the training dynamics of deep ResNets, including the emergence of fixed points, the role of regularization, and the effect of width and depth on training [S7]. Additionally, it has been shown that residual networks behave like ensembles of relatively shallow networks, which leads to a better understanding of the training dynamics of these models [S8].

Conclusion

In conclusion, residual learning is a powerful concept in deep learning that has been shown to improve the performance of various deep learning models [S1]. The use of residual connections, such as Pre-Norm and Post-Norm variants, and new methods like ReZero, can help to improve the stability and efficiency of deep neural networks [S4][S5][S6]. Further research is needed to fully understand the potential of residual learning and its applications in different fields [S7][S8].

Sources

[S1] https://aclanthology.org/2025.acl-long.1375.pdf

[S2] https://people.idsia.ch/~juergen/who-invented-residual-neural-networks.html

[S3] https://shchegrikovich.substack.com/p/where-to-put-normalization-layer

[S4] https://kindxiaoming.github.io/blog/2026/depth-1

[S5] https://arxiv.org/pdf/2003.04887

[S6] https://proceedings.mlr.press/v161/bachlechner21a/bachlechner21a.pdf

[S7] https://web.stanford.edu/~lexing/mfresnet.pdf

[S8] https://proceedings.neurips.cc/paper_files/paper/6556-residual-networks-behave-like-ensembles-of-relatively-shallow-networks.pdf